ASLAM KHAN MOHAMMED

DATA ENGINEER — AI/ML DATA PLATFORMS

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience designing, building, and operating large-scale batch, streaming, and ML-ready data platforms across AWS, Azure, and GCP. Deep hands-on expertise in Python, SQL, Scala, Apache Spark, PySpark, Kafka, and Airflow, with working exposure to Flink, Terraform, Kubernetes, and multi-cloud data services. Experienced building feature engineering pipelines, curated training datasets, and governed, high-quality data foundations that support machine learning, MLOps, and AI product development. Strong collaborator with data scientists, ML engineers, analysts, and platform teams in Agile delivery environments.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Java, PL/SQL

Cloud (AWS – Primary): EC2, S3, Glue, Redshift, EMR, Lambda, Kinesis, RDS, IAM, CloudWatch, Athena, Step Functions, SageMaker

Cloud (Exposure): Azure Data Factory, Azure Synapse, Azure Blob Storage; GCP BigQuery, Cloud Storage, Dataflow, Vertex AI

Big Data & Streaming: Apache Spark, PySpark, Apache Kafka, Apache Flink, Hadoop, HDFS, Hive, MapReduce

Orchestration & IaC: Apache Airflow, Terraform, Docker, Kubernetes, Jenkins, GitHub Actions (CI/CD)

Data Warehousing & Modeling: Snowflake, Redshift, dbt, ETL/ELT, Dimensional & Star-Schema Modeling, Databricks Lakehouse

Databases: PostgreSQL, MySQL, Oracle, MongoDB, DynamoDB, SQL Server

Data Quality & Governance: Great Expectations, Data Validation Frameworks, Data Lineage, Data Security, FHIR/HL7

ML / AI Engineering Support: Pandas, NumPy, PySpark MLlib, Scikit-learn, TensorFlow, Feature Engineering, MLflow, Feature Store Design, Vector Databases (FAISS, Pinecone), LLM/RAG Data Pipelines, MLOps

Version Control & Agile: Git, GitHub, Bitbucket, SVN, Jira, Scrum/Agile

CORE COMPETENCIES

Data Pipeline Architecture • ETL/ELT Development • Batch & Streaming Processing • ML/AI Data Enablement • Feature Engineering • Cloud Data Platforms (AWS/Azure/GCP) • Infrastructure as Code • Dimensional Data Modeling • Data Quality & Governance • CI/CD & Containerization • Agile/Scrum Delivery

PROFESSIONAL EXPERIENCE

Data Engineer | *T-Mobile, New York, NY* **Sep 2023 – Present**

- Design, build, and maintain scalable batch and near-real-time data pipelines using Python, SQL, Scala, Apache Spark/PySpark, Apache Kafka, and Apache Flink to support analytics, operational reporting, and downstream ML use cases.
- Architect and provision cloud data infrastructure on AWS (EC2, S3, Glue, EMR, Lambda, Kinesis, Redshift, RDS, DynamoDB, Athena, Step Functions) using Terraform for repeatable, version-controlled infrastructure-as-code.
- Develop and orchestrate ETL/ELT workflows in Apache Airflow that ingest, cleanse, transform, and publish data from relational, API, and event-driven sources into Redshift and Snowflake.
- Build reusable dimensional and star-schema data models and curated datasets in dbt and Snowflake, improving data accessibility for analysts, data scientists, and downstream applications.
- Design and maintain feature engineering pipelines and reusable feature sets in a feature store, partnering with data science and ML engineering teams to prepare high-quality training and inference data for production of ML models.
- Implement automated data validation and quality checks with Great Expectations, along with reconciliation, monitoring, and lineage tracking, to strengthen pipeline reliability and trust in production and ML-facing data.
- Containerize and deploy data and ML-support services with Docker and Kubernetes and manage CI/CD release pipelines through Jenkins and GitHub Actions, including MLflow-tracked pipeline components.
- Optimize Spark jobs, SQL queries, partitioning, and storage layouts to improve throughput, reduce processing latency, and control cloud costs; monitor production workloads with CloudWatch.
- Design and implement real-time feature-serving pipelines using Kinesis and Lambda to support low-latency ML inference for personalization and anomaly-detection use cases.
- Build and maintain data catalogs and metadata management practices using AWS Glue Data Catalog and Athena, improving data discoverability and self-service access for analytics and data science teams.
- Mentor junior data engineers on pipeline design, code review standards, and testing practices lead sprint planning and technical design discussions for the data platform team.
- Collaborate with product, analytics, data science, and engineering teams in Agile/Scrum environments using Jira and Git-based workflows; document architecture, technical decisions, and production support runbooks.

Environment: *Python, SQL, Scala, Apache Spark, PySpark, Apache Kafka, Apache Flink, Airflow, AWS (EC2, S3, Glue, EMR, Lambda, Kinesis, Redshift, RDS, DynamoDB, Athena, Step Functions, IAM, CloudWatch, SageMaker), Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, Snowflake, dbt, Great Expectations, MLflow, Git, Agile/JIRA.*

Data Engineer | *Tata Consultancy Services (TCS), New York, NY* **Feb 2022 – Jul 2023**

- Engineered and supported automated data pipelines using Python, SQL, Spark, and Airflow to move structured and semi-structured data from source systems into analytical environments across Azure and GCP.
- Built ingestion and transformation workflows leveraging Azure Data Factory, Azure Synapse, and Azure Blob Storage, alongside GCP BigQuery, Cloud Storage, and Dataflow, to support multi-cloud client environments.
- Developed dimensional and relational data models for reporting and analytics, including fact, dimension, staging, and curated layers, using Hive, Hadoop, and MapReduce for large-volume batch processing.
- Prepared cleaned, feature-ready datasets and reusable transformation components in dbt for downstream machine learning and analytics workloads, improving consistency across ingestion pipelines.
- Performed query tuning across PostgreSQL, Oracle, SQL Server, and MongoDB, along with pipeline troubleshooting, root-cause analysis, and operational monitoring to meet data delivery SLAs.
- Partnered with business analysts, data science teams, and client stakeholders to translate requirements into technical specifications, source-to-target mappings, and test cases.
- Migrated legacy on-premises ETL workflows to cloud-native pipelines on Azure and GCP, reducing manual intervention and improving pipeline scalability and reliability.
- Designed and implemented data security and access control policies across databases and cloud storage layers to help meet client compliance and data governance requirements.
- Authored technical documentation and knowledge-transfer materials that support onboarding of new team members and smooth client handoffs.
- Used Git-based development, Docker, code reviews, Jira tracking, and Agile ceremonies to deliver maintainable, well-documented data engineering solutions.

Environment: *Python, SQL, Apache Spark, PySpark, Airflow, dbt, BigQuery, Dataflow, Cloud Storage, Azure Data Factory, Azure Synapse, Azure Blob Storage, Snowflake, PostgreSQL, Oracle, SQL Server, MongoDB, Hive, Hadoop, Git, Docker, Jira.*

Junior Data Engineer | WIPRO*, New York, NY* **Jan 2021 – Jan 2022**

- Supported the design and maintenance of data pipeline feeding and AI Engine platforms, helping transform raw data from diverse global sources into clean, analysis-ready, ML-consumable datasets using Python, SQL, and Pandas/NumPy.
- Assisted senior data engineers in building and testing batch ingestion workflows and reusable SQL queries that supported downstream machine learning model training and analytics use cases for retail and manufacturing clients.
- Helped implement data validation and quality checks on ingested datasets and supported early-stage feature engineering and exploratory data analysis using Scikit-learn.
- Investigated data discrepancies, documented findings, and supported defect resolution and pipeline testing in close collaboration with Data Science and Product teams, using Git version control and Jira.
- Contributed to internal tooling that automated routine data pipeline monitoring and alerting tasks, reducing manual QA effort for the data engineering team.
- Built foundational familiarity with AWS and GCP services while supporting cloud infrastructure needs for AI product data pipelines.
- Participated in code reviews, technical documentation, team stand-ups, and Agile delivery activities while developing foundational data and ML-support engineering skills in a fast-paced AI product environment.

Environment: *Python, SQL, Pandas, NumPy, Scikit-learn, ETL, data validation, Git, Agile/JIRA.*

PROJECTS & CERTIFICATION

- Accident Severity Insights Pipeline: Built an end-to-end data pipeline for ingesting, cleaning, transforming, and analyzing accident datasets, using PySpark and SQL to prepare features and support ML-based severity insights and reporting dashboards.
- TalkRx: Developed a FastAPI and FHIR-oriented backend supporting structured healthcare data exchange and API-driven application workflows, with PostgreSQL as the persistence layer.
- ThermoGuardian AI: Designed a smart continuous fever-monitoring concept using distributed edge devices, real-time data aggregation, signal processing, and ML trend analysis with Scikit-learn.
- Data Quality Automation Framework: Built a reusable Great Expectations-based validation framework adopted across multiple pipelines to standardize data quality checks and reduce manual QA effort.
- Certification: AWS Certified Cloud Practitioner.

EDUCATION

Master of Science in Computer Engineering | California State University Northridge, CA
Bachelor of Engineering in Information Technology | Osmania University, Telangana